SwiftLog Technical Report

Machine Learning Track | Mission ML05: Intelligent Logistics Prediction & Supply Chain Optimization

1. Purpose

This report documents the machine-learning workflow for four logistics tasks using the supplied SwiftLog educational datasets. The workflow includes data quality review, feature engineering, three candidate algorithms per task, held-out evaluation, limited hyperparameter tuning, permutation importance, saved model bundles, and a Streamlit prototype.

2. Data preparation

Source tables were loaded from Data/Original_Datasets, text fields were standardized, exact duplicate rows were removed from processed copies, and row counts, duplicate counts, and missing cells were recorded. Related order, item, product, customer, shipment, warehouse, and return tables were joined where required.

3. Targets and feature design

Delivery delay: positive when DelayDays > 0; DelayDays and post-event delivery status are excluded from predictors. Product return: positive when an OrderID appears in Returns; order timing, city, and order-item aggregates are used. Demand forecasting: weekly product quantity is the target; lagged prior-week demand, a four-week lag, and a rolling mean of the four preceding weeks are predictors. Customer value: next-month revenue is the target; features summarize customer activity in the preceding three months.

4. Evaluation design

Classification tasks use a stratified random 80/20 split and report accuracy, precision, recall, F1, and ROC-AUC. The demand and customer-value tasks use chronological train/test splits so later periods are held out. Regression reports MAE, RMSE, and R-squared. Tuning is limited and results should be treated as experimental.

5. Model comparison

The following table reports all candidate algorithms. Selected marks the saved model; the selected row is updated to match the saved model metrics.

Problem	Algorithm	Selected	Tuned	F1	ROC_AUC	MAE	RMSE	R2	Accuracy
Delivery Delay Classification	Random Forest (tuned)	True	True	0.904	0.496				0.827
Delivery Delay Classification	Decision Tree	False	False	0.730	0.546				0.597
Delivery Delay Classification	Logistic Regression	False	False	0.666	0.484				0.527
Product Return Classification	Decision Tree	True	False	0.161	0.481				0.470
Product Return Classification	Logistic Regression	False	False	0.149	0.494				0.554
Product Return Classification	Random Forest	False	False	0.000	0.526				0.876
Product Demand Forecasting	Random Forest Regressor (tuned)	True	True			6.404	7.957	-0.028	

Problem	Algorithm	Selected	Tuned	F1	ROC_AUC	MAE	RMSE	R2	Accuracy
Product Demand Forecasting	Decision Tree Regressor	False	False			6.598	8.402	-0.146	
Product Demand Forecasting	Linear Regression	False	False			6.717	8.337	-0.129	
Customer Value Prediction (Next Month)	Linear Regression	True	False			3160.402	4082.078	0.001	
Customer Value Prediction (Next Month)	Decision Tree Regressor	False	False			3290.483	4446.409	-0.185	
Customer Value Prediction (Next Month)	Random Forest Regressor	False	False			3161.888	4136.254	-0.025	

6. Interpretation

The delivery-delay target is imbalanced, and its ROC-AUC is close to 0.5. Its high F1 should therefore not be interpreted as strong ability to distinguish delayed from on-time shipments. The return classifier also has weak discrimination. The demand and next-month customer-value models have negative held-out R-squared, indicating worse performance than a simple mean baseline on their test periods. These results are reported transparently rather than presented as successful forecasts.

7. Feature importance and artefacts

Permutation importance is saved in Models/Permutation_Importance.csv and visualized in Models/Feature_Importance.png. Classification confusion matrices are saved under Models/. The selected pipelines and their feature definitions are stored in Models/Best_Model.pkl. The Streamlit app loads this bundle for interactive demonstrations.

8. Assumptions and limitations

The delay target is defined from DelayDays because the supplied status field may disagree with the numeric delay field. A recorded return is treated as the return label. Supplier reliability is not modelled because the available tables do not directly link supplier IDs to shipments. Customer value is defined as next-month item revenue rather than lifetime value. The data covers a limited historical period and does not establish production-grade predictive performance.

9. Recommendations

Improve data capture for planned and actual delivery times; reconcile delivery status with delay days; capture supplier IDs on purchase and shipment records; maintain time-stamped inventory snapshots; define return labels consistently; collect more historical periods; evaluate models against simple baselines; and pilot any predictions with human review before operational use.

10. Reproduction

Install dependencies from App/requirements.txt or the README instructions, then run python run_swiftlog_project.py from the project root. The script regenerates processed datasets, model comparisons, saved model bundles, evaluation figures, reports, and the presentation. The notebook reviews the outputs.